

jQuery and AJAX Notes

1. Including jQuery

To use jQuery in your webpage, include the following line inside the `<head>` tag:

```
<script src="https://code.jquery.com/jquery-3.6.0.min.js"></script>
```

2. Checking if jQuery is Loaded

You can verify that jQuery is properly loaded using:

```
$(function(){  
    alert("jQuery is loaded and ready!");  
});
```

- `$(function(){ ... })` runs when the HTML document is fully loaded.

3. Handling Button Click Events

To perform an action when a button is clicked:

```
$("#btn").click(function(){  
    alert("Button clicked!");  
});
```

- `#btn` refers to an element with `id="btn"`.

4. Mouse Events

```
$("#box").mouseenter(function(){  
    $(this).css("background-color", "yellow");  
});
```

- Changes the box color when the mouse enters.

```
$("#box").mouseleave(function(){  
    $(this).css("background-color", "white");  
});
```

- Changes it back when the mouse leaves.

5. Keyboard Events

```
$("#inputBox").keypress(function(){  
    console.log("Key pressed!");  
});
```

```
});
```

- Detects when any key is pressed inside an input field.

6. Show and Hide Elements

```
$("#btn").click(function(){  
    $("#content").toggle();  
});
```

- `toggle()` shows or hides an element each time the button is clicked.

7. Fade Effects

```
$("#fadeInBtn").click(function(){  
    $("#box").fadeIn();  
});  
  
$("#fadeOutBtn").click(function(){  
    $("#box").fadeOut();  
});
```

- `fadeIn()` makes an element appear gradually.
- `fadeOut()` makes it disappear gradually.

8. Slide Effect

```
$("#slideBtn").click(function(){  
    $("#panel").slideToggle();  
});
```

- `slideToggle()` slides an element up or down smoothly.

9. Changing Text and HTML

```
$("#btn").click(function(){  
    $("#heading").text("New Text");  
    $("#content").html("<strong>New HTML Content</strong>");  
});
```

- `.text()` changes only text content.
- `.html()` replaces the HTML structure inside an element.

10. Changing Attributes

```
$("#img").attr("src", "new-image.jpg");
```

- `.attr()` changes an attribute value (for example, updating an image source).

11. Working with Classes

```
$("#box").addClass("highlight");  
$("#box").removeClass("highlight");  
$("#box").toggleClass("highlight");
```

- `.addClass()` adds a CSS class.
- `.removeClass()` removes it.
- `.toggleClass()` switches between adding and removing the class.

12. Traversing Elements

```
$("#child").parent().css("border", "2px solid red");  
$("#parent").children().css("color", "blue");  
$("#sibling").siblings().hide();
```

- `.parent()` selects the parent element.
- `.children()` selects direct child elements.
- `.siblings()` selects all elements at the same level.

13. Chaining Methods

How multiple jQuery methods can be combined:

```
$("#box").css("color", "blue").slideUp(1000).slideDown(1000);
```

- Chaining improves code readability and performance.

14. Getting and Setting Values

Very common in form handling:

```
let name = $("#nameInput").val(); // get value  
$("#nameInput").val("Wasee"); // set value
```

15. Removing and Appending Elements

Useful for dynamic pages:

```
$("#list").append("<li>New Item</li>");  
$("#listItem").remove();
```

- `.append()` adds content at the end of an element.
- `.remove()` deletes elements from the DOM.

16. AJAX with jQuery

```
$("#searchBox").keyup(function(){  
    let query = $(this).val();  
    if(query.length > 2) {  
        $.ajax({  
            url: "someURL",  
            method: "GET",  
            data: { query: query },  
            success: function(response){  
                $("#results").html(response);  
            }  
        });  
    } else {  
        $("#results").html("");  
    }  
});
```

Explanation:

- Runs whenever the user types in the search box.
- If the input has more than 2 characters:
 - Sends a `GET` request to the server (`someURL`) with the typed text as `query` .
 - On success, updates the `#results` section with the response.
- If the input is empty, it clears the results.

Examples

Example 1:

A jQuery code so that when the mouse pointer is moved over a paragraph, its background becomes red. When the mouse leaves, the color should return to white.

Answer:

```
$("p").mouseenter(function(){  
    $(this).css("background-color", "red");  
});
```

```
$("p").mouseleave(function(){
    $(this).css("background-color", "white");
});
```

Example 2:

A jQuery code that hides all paragraph elements when a button with ID `"hideBtn"` is pressed.

Answer:

```
$("#hideBtn").click(function(){
    $("p").hide();
});
```

Example 3:

A jQuery code that displays all paragraph elements again when a button with ID `"showBtn"` is clicked. (5)

Answer:

```
$("#showBtn").click(function(){
    $("p").show();
});
```

Reference:

Prepared by **Wasee Ur Rehman**

 *Web Programming Notes — jQuery & AJAX Summary*

 <https://linkedin.com/in/wasee-ur-rehman>